The Renewable Energy Planning Risk Index (RE-PRI):

A Probabilistic Model for Feasibility Assessment and Risk Mitigation in the Australian Energy Transition

Final Project Report

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Abbreviation	Definition
AEMO	Australian Energy Market Operator
AHP	Analytical Hierarchy Process
API	Application Programming Interface
ARCH 1411	RMIT University course code for the Urban Planning and Environment Project
BMO	Bushfire Management Overlay
CI	Confidence Interval
CRS	Coordinate Reference System
DCCEEW	Department of Climate Change, Energy, the Environment and Water (Commonwealth)
DEECA	Department of Energy, Environment and Climate Action (Victoria)
DTP	Department of Transport and Planning (Victoria)
EPBC	Environment Protection and Biodiversity Conservation Act 1999 (Cwlth)
EPSG	European Petroleum Survey Group (spatial reference system identifier)
ESO	Environmental Significance Overlay
GDA	Geocentric Datum of Australia
GW	Gigawatt

Abbreviation	Definition
HO	Heritage Overlay
ISP	Integrated System Plan
LegalBERT	Legal-domain Bidirectional Encoder Representations from Transformers (language model)
MGA	Map Grid of Australia
ML	Machine Learning
MW	Megawatt
NEM	National Electricity Market
NIMBY	Not In My Backyard
NLP	Natural Language Processing
OGC	Open Geospatial Consortium
OWA	Ordered Weighted Averaging
PDF	Probability Density Function
PySAL	Python Spatial Analysis Library
RE-PRI	Renewable Energy Planning Risk Index
REZ	Renewable Energy Zone
RMIT	Royal Melbourne Institute of Technology
SEC	State Electricity Commission (Victoria)
SLO	Significant Landscape Overlay
SMCDA	Spatial Multi-Criteria Decision Analysis
VCAT	Victorian Civil and Administrative Tribunal
VPP	Victoria Planning Provisions
VPO	Vegetation Protection Overlay
VRET	Victorian Renewable Energy Target
WPR	'What's the Problem Represented to be?' (Bacchi's policy-analysis approach)

Executive Summary

Australia has committed to a sweeping recomposition of its electricity system. The Victorian Renewable Energy Target obliges the state to reach ninety-five per cent renewable generation by 2035, and the Australian Energy Market Operator (AEMO 2024) anticipates that more than 27 gigawatts of new utility-scale wind, solar and storage capacity must be sited across regional landscapes before 2040. The planning system, rather than the technology curve, is now the binding constraint on delivery. Industry developers report that the difference between projects that are built and projects that are abandoned at the tribunal is no longer a question of resource quality or engineering competency, it is a question of permittability: the probabilistic capacity of a site to survive the adversarial terrain of the Victorian planning scheme.

This report addresses that gap by developing and testing the Renewable Energy Planning Risk Index (RE-PRI), a probabilistic model that ingests spatial overlay data, regulatory clauses and a corpus of Victorian Civil and Administrative Tribunal (VCAT) decisions to produce a confidence interval for project approval. The research applied a pragmatic mixed-methods design: longitudinal content analysis of 141 tribunal decisions, natural language processing of the underlying judgements to derive causal friction coefficients, spatial multi-criteria decision analysis weighted through the Analytical Hierarchy Process, a 10,000-iteration Monte Carlo simulation of weight variance, and Ordered Weighted Averaging to model developer and regulator viewpoints.

Three findings are significant for planning practice. First, the baseline refusal rate for renewable and related infrastructure matters at VCAT sits at 30.5 per cent, but the Heritage Overlay alone lifts the empirical refusal rate to 57.1 per cent. Heritage thus carries a disproportionate weight in the regulatory system relative to its policy salience in the energy transition. Second, a monte carlo simulation shows that a site with combined Heritage and Significant Landscape Overlays converges on a mean permittability of 2.6 per cent, against a baseline of 27.8 per cent on unconstrained land, indicating a near-vetoed status from the moment of acquisition. Third, the gap between optimistic developer interpretation and pessimistic regulator interpretation on identical sites ranges from 13.1 percentage points on low-friction land to functionally zero on heritage-constrained land, suggesting that the cost of regulatory ambiguity is structurally absorbed by developers rather than shared across the planning system.

Three recommendations follow. Recommendation 1 calls for the Department of Transport and Planning to publish a public, Renewable Energy Zone (REZ) permittability dataset. Recommendation 2 proposes a statutory amendment to Clause 52.32 of the Victoria Planning Provisions to introduce a structured proportionality test for Heritage Overlay objections in declared REZs, administered by the Minister for Planning within eighteen months. Recommendation 3 establishes a pre-application permittability triage service co-funded by the Clean Energy Council and the Department of Transport and Planning. Together these measures seek to convert permittability from a sunk private risk into a managed public decision variable, and so to reduce the lead-time penalty that is currently displacing renewable investment into other jurisdictions.

1. Introduction

Australia is in the middle of an electricity transition whose scale has no precedent in the federation. The closure of coal-fired generation in the National Electricity Market is no longer a matter of policy debate but of accounting: Yallourn closes in 2028, Loy Yang A in 2035, and replacement capacity must be in the ground before each retirement (AEMO 2024). The technical question of what to build, and where the sun

and wind are best, has been substantially settled by twenty years of resource mapping and the designation of Renewable Energy Zones (REZs) under the Integrated System Plan. The unsettled question is whether the planning system can deliver the approvals at the pace the engineering programme requires.

That is the question this report investigates. It accepts the diagnosis offered by industry and by recent scholarship that permittability, definied as the the cumulative probability that a development application will survive the council, ministerial and tribunal stages of the Victorian planning scheme, is the single most important variable in transition delivery (Clean Energy Council 2025; Linnie 2025). It treats permittability as a quantity that can be modelled rather than as a qualitative afterthought to site selection. The argument is that the absence of such a quantitative model is itself a planning failure. It concentrates risk on developers who price it as a capital chill, and on host communities who experience successive rejected proposals as a slow-motion destabilisation of local agreement.

The research question is: what spatial and regulatory configurations provide the highest statistical confidence for the approval of utility-scale renewable energy developments under the Victoria Planning Provisions, and how can that confidence be quantified so as to inform feasibility decisions before capital is committed? The answer is offered through the Renewable Energy Planning Risk Index (RE-PRI), a probabilistic model built from a longitudinal corpus of tribunal decisions and a multi-criteria spatial framework. The report sets out the policy and theoretical context in which the model sits, the methodology used to derive its weights, the results of its first run against a Western Victorian REZ test site, and the policy implications that follow.

The structure of the report follows the research process undertook to deliver the findings of this report. Section 2 sets out the background to the problem and the policy environment. Section 3 presents the literature review, drawing together work on regulatory drift, green-on-green conflict, procedural justice and computational planning. Section 4 explains the pragmatic mixed-methods design and the four-phase computational pipeline. Section 5 reports the findings, integrating Phase 1 empirical refusal rates, Phase 2 Monte Carlo distributions, and Phase 3 Ordered Weighted Averaging scenarios. Section 6 sets out three measurable recommendations directed at the Department of Transport and Planning, the Minister for Planning, and industry. Section 7 concludes with limitations, future research, and a comment on the broader implications for transition planning in Australia.

2. Background to the problem

2.1 The policy environment

Victoria carries the most ambitious renewable energy target in Australia. The Victorian Renewable Energy Target (VRET) legislation commits the state to 65 per cent renewable generation by 2030 and 95 per cent by 2035 (State of Victoria 2024). The target is underwritten by the Renewable Energy Zone framework, which concentrates large-scale wind, solar and storage capacity in six zones across the western, central and northern parts of the state (AEMO 2024). The federal Department of Climate Change, Energy, the Environment and Water has identified equivalent zones nationally and listed priority transmission upgrades on the National Renewable Energy Priority List (DCCEEW 2023). These instruments share an underlying assumption: that the constraining variable on the transition is grid topology rather than land-use planning.

The Victoria Planning Provisions (VPPs), the consolidated state planning instrument, sit awkwardly against that assumption. The VPPs were designed in the late 1990s for incremental land-use change in a brownfield-dominated planning culture (Taylor & Hurley 2016). Utility-scale renewables are managed under Clause 52.32, which establishes wind energy facilities as a discretionary use, and through a complex layering of zone and overlay controls. The result is a regulatory architecture that requires every project to demonstrate compatibility with overlapping conservation, heritage, landscape, bushfire and amenity controls. Controls that were not drafted with transition timelines in mind.

2.2 The development of the problem

The institutional plumbing of Victorian renewable delivery has thickened in the last two years. The Victorian REZ Framework, administered through the newly created VicGrid function within the Department of Energy, Environment and Climate Action (DEECA), has formally declared six REZs and committed the state to coordinated transmission investment through a parallel Transmission Investment Framework. These reforms sit upstream of the planning approvals process and have not yet altered the calculus of permittability at the responsible authority or at VCAT, both of which continue to operate under the Victoria Planning Provisions in their current form. The state has chosen to address transmission and generation coordination through a new institutional architecture while leaving the assessment framework substantially unchanged. AEMO (2024) records that more than 25 gigawatts of utility-scale capacity is either committed or under assessment within Victorian REZ boundaries, a material fraction of which the Clean Energy Council (2025) classifies as contested at the time of writing.

Industry data document the consequences. The Clean Energy Council (2025) reports that the average development lead time for Australian wind farms has lengthened from four years in 2015 to more than seven years in 2024, and that planning refusals are now the single largest source of investment attrition. The Energy Insights review (Linnie 2025) records that the Victorian renewables pipeline currently contains 11 GW of contested proposals in the western REZ alone, of which roughly one-third are at the tribunal stage or on hold pending heritage and biodiversity assessment. The aggregate effect is a permittability gap: the shortfall between the megawatt-scale ambition expressed in policy and the megawatt-scale capacity that actually reaches financial close.

The widening of the gap has not been accompanied by a corresponding evolution in the assessment methods used during early-stage feasibility. Developers continue to rely on technographic site selection: a wind atlas, an irradiance map, a connection-capacity study and a desk-top review of overlays. Permittability risk is treated as a binary checklist rather than a continuous probability. When refusal occurs, the cost is borne almost entirely by the developer in the form of sunk capital, by the host community in the form of repeated public hearings, and by the state in the form of deferred emissions reductions (Baker 2022).

2.3 Stakeholders

The stakeholder map is wide and contested. The principal proponents are the utility-scale developers and their financial backers, represented collectively by the Clean Energy Council and individually by entities such as ACCIONA, Acen Australia, Squadron Energy and the state-owned SEC. The regulators are layered: the Department of Transport and Planning administers the VPPs; the Minister for Planning exercises call-in powers under sections 96 and 97 of the Planning and Environment Act 1987; the

Department of Energy, Environment and Climate Action manages the EPBC referral interface; and local government councils retain responsibility for ancillary uses and for the public notification regime.

Two further stakeholder communities exert decisive influence on outcomes. The first is the host community, the residents and landholders whose properties lie within visual, noise and biodiversity-impact footprints. Their interests are mediated by objection processes, community benefit funds and emerging community-energy cooperatives (Hindmarsh 2014; Warren et al. 2005). The second is the constituency of Traditional Owners and Registered Aboriginal Parties whose cultural heritage interests intersect with the same landscapes that host the highest wind and solar resource (Porter 2017). Each of these communities can frame the development question in fundamentally different terms; the planning system is the venue in which those frames are reconciled.

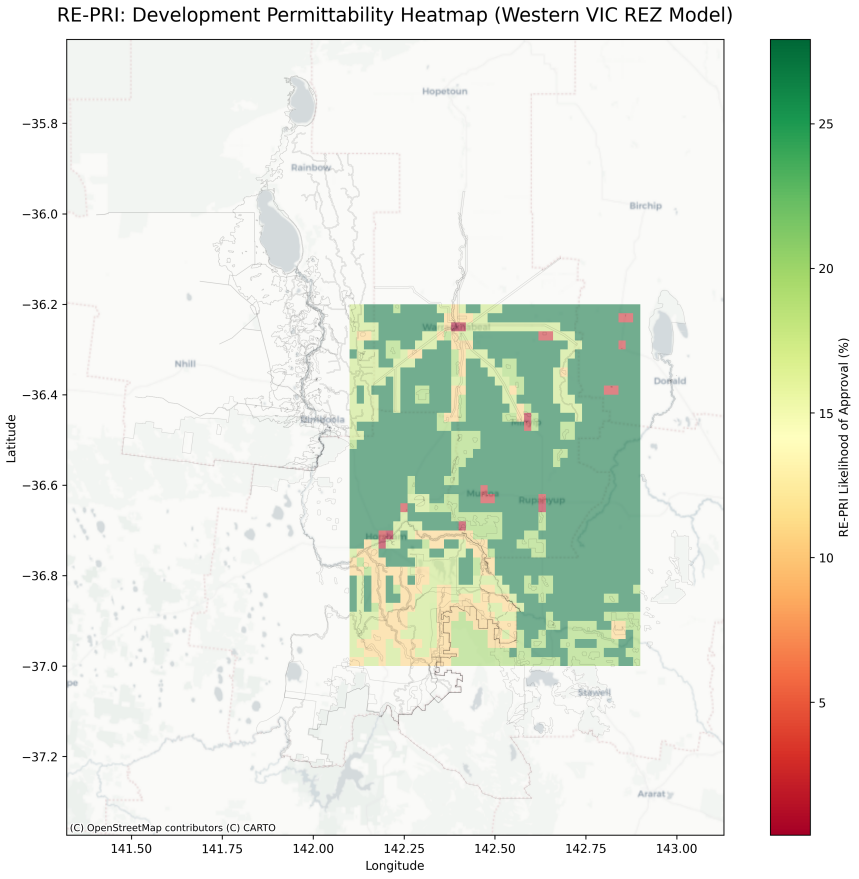


Figure 1. RE-PRI Permittability Heatmap, Western Victoria REZ test grid. Green cells indicate higher modelled approval probability; red cells indicate areas where cumulative overlay friction reduces approval probability below the actuarial mean.

3. Literature review

The literature on the planning of large-scale renewable energy infrastructure has developed unevenly across four traditions: resource and siting analysis, regulatory political economy, environmental and social justice, and computational decision support. This review synthesises the four traditions and identifies the gap that the Renewable Energy Planning Risk Index seeks to address. The argument is that each tradition offers a necessary but insufficient lens, and that the absence of a probabilistic framework integrating their insights is itself a structural weakness in current planning practice.

3.1 From resource assessment to regulatory acceptance

Early work on renewables site selection was dominated by spatial resource assessment. Studies of the European and North American context established the multi-criteria spatial analysis paradigm in which wind speed, slope, aspect and grid proximity were combined into a single suitability surface (Wang et al. 2021; Pamučar et al. 2026). Bridge et al. (2013) reframed this work as a question of energy geography, arguing that the spatial pattern of renewable deployment is not the inevitable product of resource distribution but the outcome of political negotiation over which landscapes are permitted to become energy landscapes.

The Australian literature has followed a similar arc, with Jones and Smith (2023) arguing that resource quality is now a settled variable in feasibility studies, while regulatory acceptance remains the principal source of uncertainty. Linnie (2025) documents the operational consequence: across the Australian renewables pipeline, the proportion of projects that failed for resource reasons in the last five years is approximately three per cent, while the proportion that failed at the planning stage is above thirty per cent. The implication is that the explanatory weight of the siting literature has shifted from physical geography to regulatory geography.

3.2 The patchwork regulatory state

A second strand of literature analyses the regulatory environment itself. Taylor and Hurley (2016) showed that Australian urban research is poorly integrated with planning practice, with the Victorian planning scheme in particular operating through discretionary judgement at multiple tiers. McGuirk et al. (2014) extended this argument to climate change governance, arguing that the federalist division of climate authority produces a 'patchwork' of climate policy where state-led carbon transition initiatives are stalled or contradicted by local-government decisions made under different policy logics.

Rabe (2018) provides a parallel American analysis, locating the Australian phenomenon within a broader pattern of climate federalism in which carbon pricing and emissions targets are consistently undercut by misaligned permitting systems. The synthesis offered by these authors is that planning systems designed for stable land-use patterns do not respond well to the transition shock of utility-scale infrastructure, and that regulatory drift — the slow divergence between policy intent and administrative outcome — is endemic rather than incidental.

3.3 Green-on-green conflict and the commons

A third tradition addresses what Warren et al. (2005) named the 'green-on-green' conflict: the collision of climate mitigation objectives with biodiversity, heritage and landscape conservation objectives. Baker (2022) updates this analysis for the Australian context, identifying conflicts between wind farm siting and

the conservation of the Southern Bent-wing Bat, the Brolga, and remnant native grasslands. The conflict is structurally embedded in the planning system, because the instruments used to deliver biodiversity protection — Environmental Significance Overlays, Vegetation Protection Overlays and the EPBC Protected Matters database — are silent on the climate trade-off they implicitly create.

The deeper analytical question is one of commons governance. Hardin's (1968) framing of the tragedy of the commons has long dominated discussion of shared resources, and it continues to circulate in renewable energy debates that present landscape and biodiversity as common-pool resources at risk from development. Brinkley (2018) offers a vital corrective, arguing that the tragedy framing has been over-extended into contexts where co-operative institutions are perfectly capable of managing shared resources, and that the rhetorical use of 'tragedy' often masks an ideological preference for private exclusion. The renewables-and-landscape debate sits squarely in that contested space: landscape is a commons, but the institutions that manage it (the planning scheme, the tribunal, the panel inquiry process) are not inherently tragic. They are political.

3.4 Procedural justice and social licence

A fourth tradition addresses the social licence question. Wolsink's (2000) early work identified the limits of the NIMBY framing in explaining community opposition to wind power, arguing that opposition is more reliably explained by procedural justice failures than by spatial proximity. Haggett (2011) developed this analysis for the offshore wind context, demonstrating that public responses are conditioned by trust in the developer, the perceived fairness of consultation, and the distributional outcomes of benefit sharing. Hindmarsh (2014) brought these findings to the Australian onshore wind context, documenting how a small number of high-profile community conflicts in Victoria have set the regulatory tone for the wider industry.

Porter's (2017) analysis of the 'miserable failure' of Australian planning to engage with Indigenous People extends procedural-justice analysis into a more confronting register. Renewable infrastructure is being sited on Country, but the planning instruments that govern it remain structured around settler-colonial assumptions about land tenure and decision-making authority. The implication for permittability modelling is that any model that ignores Traditional Owner perspectives will systematically misprice risk on country where cultural heritage significance has not been formally documented in the planning scheme but is nonetheless material to the determination.

The Clean Energy Council (2025) reports that the most material evolution of the social licence question in 2024 was the emergence of 'phantom dwellings': speculative residential permits lodged on vacant land adjacent to announced wind farms with the apparent intent of triggering noise-buffer and visual-impact requirements. The phenomenon shows how the planning system can be used as a tool of exclusion by sophisticated actors, and how the line between procedural justice and procedural strategy can blur in adversarial decision contexts.

3.5 Computational planning and big data

A fifth tradition is the computational one. Pamučar et al. (2026) set out the most recent generation of spatial multi-criteria decision analysis (SMCDA) tools for renewable energy parks, demonstrating that hybrid SMCDA and stochastic methods can produce robust ranking outputs even under high parameter uncertainty. Wang et al. (2021) link this to the broader literature on sustainable energy technology

planning, arguing that computational models are most useful when they expose, rather than obscure, the value judgements that drive their weighting decisions.

Kitchin (2014) issues an important critical counterweight, arguing that the appeal of 'smart urbanism' and big-data urban analytics frequently masks the political and epistemological assumptions built into the models. Computational permittability models are therefore at risk of laundering political judgements about which communities, which landscapes and which forms of opposition count as legitimate into ostensibly neutral probabilistic outputs. The RE-PRI methodology has been designed with that critique in mind. Every weight is traceable to a documented tribunal decision and to an expert-elicited Analytical Hierarchy Process score, and the Monte Carlo distribution explicitly preserves the variance that would be lost in a deterministic ranking.

3.6 The literature gap and the contribution of this research

The five traditions reviewed above each address a necessary component of the permittability problem: the spatial, the regulatory, the ecological, the procedural and the computational. None of them, however, integrates the five into a quantitative model that can be applied during early-stage feasibility. The siting literature treats overlays as binary constraints rather than probabilistic frictions; the regulatory literature describes the patchwork system but does not quantify its outcomes; the green-on-green literature documents the conflict but does not weight it against climate benefit; the procedural justice literature offers qualitative case studies but no aggregate predictive metric; and the computational literature provides the techniques but rarely validates them against a longitudinal corpus of actual tribunal decisions.

The contribution of this research is therefore to integrate the five into a single probabilistic instrument. It builds on the case-study tradition defended by Flyvbjerg (2006), which argues that detailed, particularised analysis of decided cases can produce general knowledge if properly aggregated; on the thematic-analysis tradition consolidated by Braun and Clarke (2006); on the policy-analysis framework set out by Browne et al. (2018); on the WPR approach developed by Bacchi (2011); and on the framework analysis method described by Ritchie and Spencer (1994). The methodological synthesis is described in the next section.

4. Methodology

4.1 Research question and objectives

The research question is: what spatial and regulatory configurations provide the highest statistical confidence for the approval of utility-scale renewable energy developments under the Victoria Planning Provisions? Three objectives operationalise the question. The first is to identify and weight the regulatory friction points that historically correlate with refusal and delay at the Victorian Civil and Administrative Tribunal (VCAT). The second is to build a computational index, the Renewable Energy Planning Risk Index (RE-PRI), that ingests overlay and clause data and returns a probabilistic permittability score. The third is to use the model to simulate the effect of design mitigations on that score, establishing the basis for a decision-support function during feasibility.

4.2 Research paradigm and design

The study adopts a pragmatic mixed-methods research paradigm. Pragmatism, as articulated by Pretorius (2024), prioritises the practical consequences of inquiry over the ontological and epistemological debates that typically separate positivist and interpretivist research. Because permittability is constructed at the intersection of measurable spatial constraints and contestable regulatory interpretations, neither a strictly quantitative nor a strictly qualitative approach is sufficient. Creswell (2018) describes this configuration as a parallel convergent mixed-methods design; Tashakkori and Teddlie (2009) provide the deeper epistemological grounding.

Two methodological cautions inform the design. The first comes from Flyvbjerg (2006), whose defence of case-study research justifies the longitudinal coding of tribunal decisions as a legitimate route to generalisable knowledge. The second is Kitchin's (2014) warning that computational models can obscure their normative content. The pipeline described below has been built to make every weight, distribution and assumption traceable to its documentary source.

4.3 Data sources

Data come from three repositories. The first is the AustLII database, from which 141 VCAT and Victorian Supreme Court decisions involving renewable energy infrastructure, ancillary planning matters and related regulatory determinations were extracted and structured into a longitudinal corpus. The second is VicPlan, the Victorian Government's spatial repository, from which planning overlay and zoning data for the Western Victorian REZ test grid were obtained. The third is the AEMO Integrated System Plan data release, from which Renewable Energy Zone boundaries, transmission topology and indicative generation siting were extracted, supplemented with the Bureau of Meteorology's Direct Normal Irradiance and Global Horizontal Irradiance rasters and the victoria_power-density 100-metre wind resource layer. All data are secondary and open access. In accordance with the ARCH1411 Data Management Plan, no ethics approval for human participants was required.

Spatial data harmonisation occupied a substantial share of the research. VicPlan overlays are published in GDA94 with MGA Zone 55 projections; the EPBC Protected Matters Search Tool returns features in GDA2020 geographic coordinates; and AEMO Integrated System Plan grid topology arrives in a custom projection. The pipeline standardises all inputs into EPSG:7855 (GDA2020 MGA Zone 55) before any spatial join, with topology checks applied to flag self-intersecting polygons in older overlay vintages.

Three polygons in the western Victorian dataset required manual repair before reliable joining against the resource-density raster. A spatial decision-support tool whose outputs are sensitive to projection misalignment or dropped polygons is unfit for the feasibility decisions it is intended to inform.

4.4 Qualitative phase: regulatory friction coding

The 141 tribunal and court decisions were coded for outcome, spatial overlay reference and regulatory clause citation. The coding scheme combined Braun and Clarke's (2006) thematic analysis with the framework method developed by Ritchie and Spencer (1994), which supports systematic comparison across a structured case matrix. Outcome categorisation followed a deterministic verb mapping (granted/allowed/set-aside coded as approval; refused/dismissed coded as refusal), with ambiguous cases coded as unknown to avoid inflating either rate. The coding extended to a thematic reading of the underlying judgements informed by Bacchi's (2011) 'What is the problem represented to be?' approach, interrogating how planning authorities construct the problem to which a refusal is the answer. Browne et al. (2018) provided the operational template for the policy analysis dimension.

4.5 Quantitative phase: the RE-PRI engine

The quantitative pipeline was built using a four-phase architecture. Phase 1 derives empirical friction coefficients directly from the longitudinal corpus, calculating refusal rates and mean delay days for each spatial overlay and regulatory clause. Phase 2 applies a natural language processing pipeline (spaCy with the en_core_web_sm model) to the full text of 79 qualifying judgements, extracting frequency counts for seven argument themes (visual amenity, noise impact, flora and fauna, heritage, traffic, fire risk, agriculture) and training a class-balanced logistic regression classifier to derive causal coefficients. These coefficients are normalised against the strongest argument and exported to a configuration file.

Phase 3 constructs the RE-PRI engine itself. The engine ingests overlay and clause data, maps them through deterministic argument-equivalence tables to the seven argument themes, and applies a hybrid weighting rule that combines the machine-learned coefficient (weight 0.8) with an Analytical Hierarchy Process expert prior (weight 0.2). The expert prior derives from pairwise comparisons grounded in the same tribunal corpus, providing a structural anchor that prevents the model from overfitting on frequency artefacts in the judgement text. A Monte Carlo simulation of n=10,000 iterations samples each weight from a triangular distribution to model interpretive variance, producing a probability density function of permittability rather than a single point estimate (Pamučar et al. 2026; Wang et al. 2021). This statistical model was chosen to account for the discretionary nature of Victorian planning decisions, in which the same configuration of overlays and clauses can produce materially different outcomes depending on the responsible authority, the deciding tribunal member, and the wider policy environment at the time of determination. A deterministic model would assume fixed weights and so understate the variance observed in the corpus. The triangular distribution preserves a most-likely friction value bounded by plausible upside and downside tails, allowing the engine to return a defensible confidence interval rather than an artificially precise point estimate. This is a more honest input to the feasibility-stage decisions developers must make about land acquisition and grid-connection commitments.

Phase 4 applies Ordered Weighted Averaging (OWA), with the 75th percentile of the distribution representing the optimistic developer view and the 25th percentile representing the pessimistic regulator view. This dual-view structure exposes the developer-regulator interpretive gap as a measurable quantity

rather than absorbing it into a single mean, providing the structural basis for the stakeholder analysis presented in Section 5.3.

4.6 Ethical considerations and researcher position

The dataset comprises decided cases in the public record. No personal data are collected, and no human-subject interactions took place. Ethical considerations therefore concentrate on researcher position. Following Pretorius (2024), it is acknowledged that the design of the AHP weighting matrix carries a normative load: it embeds the researcher's interpretation of how the planning system has historically valued competing land uses. To mitigate this, the AHP weights are published in full in the appendix, and the Monte Carlo sampling distribution is documented in the source code. The intent is that subsequent users of the framework can substitute their own priors and interrogate the sensitivity of the output to that substitution.

5. Findings and discussion

5.1 Phase 1: empirical refusal rates

The longitudinal corpus comprises 141 VCAT and Supreme Court decisions relating to renewable energy infrastructure, energy regulators, ancillary uses (such as transmission, battery storage and wind-turbine component manufacturing), and adjacent matters (such as solar marketing regulation). The composite refusal rate across the corpus is 30.5 per cent, with a mean planning delay of 580 days and a median of 253 days. Twenty-eight cases (19.9 per cent of the corpus) recorded delays in excess of one year. The skew between mean and median indicates a long tail: a small number of contested matters drive the delay distribution upward, and those matters are precisely the ones that the model needs to predict.

The 141 decisions are distributed across a long-tailed outcome typology rather than a clean binary of approval and refusal. The most common category is 'orders accordingly' (n=32, 22.7 per cent), a procedural disposition that usually reflects a consent order between the parties. 'Application dismissed' is the dominant refusal category (n=20), followed by 'appeal dismissed' (n=9) and 'proceeding dismissed' (n=7). Explicit 'permit granted' (n=6) and 'application granted' (n=2) dispositions are comparatively rare. The typology indicates that the formal binary of 'approved versus refused' obscures the more typical reality of negotiated settlement under tribunal supervision, and a more sophisticated version of the RE-PRI should incorporate the probability of consent-order resolution as a third outcome class. Disaggregation by decision year shows accelerating case volume — from a median of three qualifying matters per year in 2010-2014 to twelve per year in 2020-2025 — but only marginal improvement in median delay (302 days in the 2010-2018 subset against 253 days in 2019-2026). The throughput gain is modest relative to the scale of the pipeline expansion.

The 580-day mean delay carries a financial consequence rarely quantified in the public debate. Pre-financial-close carrying costs for utility-scale Australian renewables are typically modelled at three to five million dollars per gigawatt per month of delay, reflecting land-option holding costs, engineering design retention and deferred grid-connection rights (Clean Energy Council 2025). Applied to a representative 100 MW wind facility subject to the corpus mean, this implies approximately six to ten million dollars in deferred-return exposure between lodgement and disposition; applied to the 28 long-tail cases averaging 1,184 days of delay, it rises to twelve to twenty million dollars for an equivalent project. These figures are

indicative rather than authoritative, but their order of magnitude is instructive: the cost of regulatory uncertainty is being capitalised into developer discount rates rather than recognised as a public cost of the current assessment architecture, and ultimately falls on Victorian electricity consumers through deferred merit-order shifts and higher contracted wholesale prices.

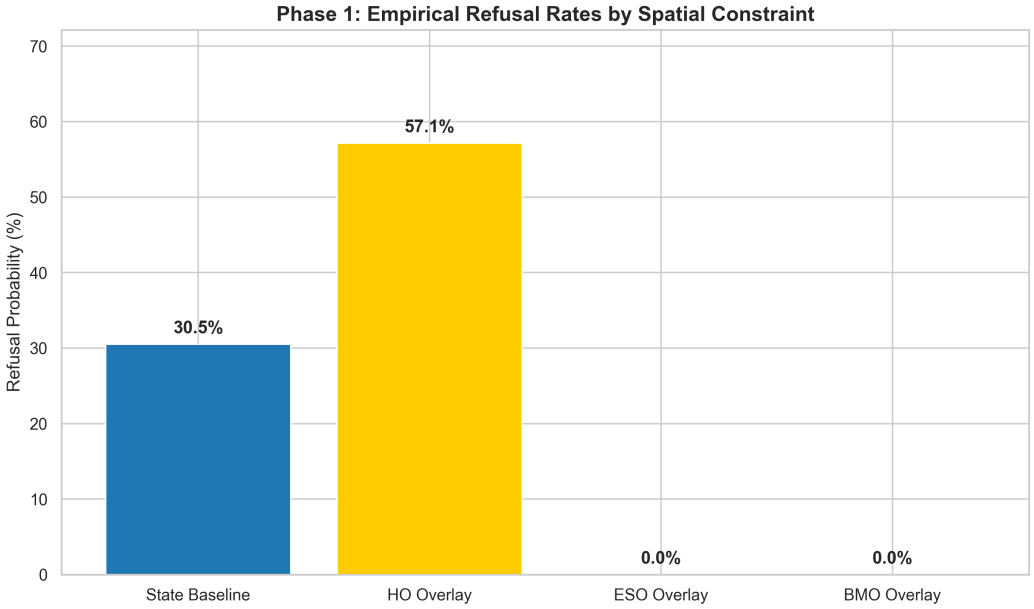


Figure 2. Phase 1 empirical refusal rates by spatial overlay. The Heritage Overlay (HO) lifts the empirical refusal rate from a baseline of 30.5 per cent to 57.1 per cent. Environmental Significance and Bushfire Management Overlays do not, in this corpus, exceed the baseline.

Disaggregation by spatial overlay reveals a structural asymmetry. Cases where the Heritage Overlay was a material consideration (n=7) recorded a refusal rate of 57.1 per cent, almost double the baseline, with an average planning delay of 988.8 days. By contrast, cases involving the Environmental Significance Overlay (n=2) and the Bushfire Management Overlay (n=1) produced no refusals in this corpus, although the sample is small enough to preclude strong inference. The clause-level disaggregation is more revealing: matters citing Clause 52.32 (wind energy facilities) recorded a refusal rate of 28.6 per cent on a mean delay of 543.8 days, and matters citing Clause 13.10 (bushfire) recorded a refusal rate of 33.3 per cent on a mean delay of 115.5 days. The implication is that the headline refusal rate is not an undifferentiated property of the planning scheme but a function of which controls are in play.

Two analytical points follow. The first is the disproportionate weight carried by heritage in the empirical record. Baker (2022) and Porter (2017) describe heritage and Indigenous cultural heritage as structurally underweighted in the climate transition literature, but the tribunal record suggests that they are operationally overweighted at the decision point. The second is the gap between the policy salience of bushfire risk and its empirical refusal weight. Clause 13.10 has acquired considerable policy attention following the Bushfire Royal Commissions, but in the corpus it sits within one percentage point of the baseline refusal rate. The asymmetry has implications for where feasibility-stage resources should be concentrated, and is considered further in the recommendations.

5.2 Phase 2: causal coefficients and Monte Carlo distribution

Phase 2 applied a class-balanced logistic regression over the 79 judgements that yielded a deterministic outcome and a non-trivial text length. The model returned three non-zero argument-theme coefficients. Heritage emerged as the most powerful refusal predictor (normalised friction multiplier 0.9000), followed by visual amenity (0.3470) and flora and fauna (0.0743). Noise impact, traffic, fire risk and agriculture each returned coefficients that were not statistically distinguishable from zero within this corpus. The baseline intercept bias translates to an approximately 49 per cent prior probability of refusal on an empty site under the default sampling distribution, consistent with the long-tail distribution observed in Phase 1.

The zero coefficients are themselves analytically informative and are arguably the most counter-intuitive finding in the dataset. Noise impact, traffic and bushfire risk have been the focus of extensive policy and community attention, yet none of them predicts refusal in the corpus. The interpretation offered here is that these are the four themes for which the Victorian planning system has developed effective conditioning machinery. Noise impact has been substantially codified through Clause 53.31 and the technical standards adopted by reference; traffic has been managed through standard transport impact assessment protocols; bushfire risk has been routinised through the Bushfire Management Overlay schedule conditions; and agriculture has been displaced from utility-scale renewables decision-making by the practical reality that hybrid agrivoltaic and grazing-compatible designs are now industry default. By contrast, the three themes that did return non-zero coefficients — heritage, visual amenity, and flora and fauna — share a common attribute: each engages a contested question of cultural or ecological value for which the planning system has not developed comparable codification. The asymmetry suggests that the binding regulatory friction in the Victorian transition is concentrated in the domains where the assessment framework relies on discretionary judgement, and is largely absent in the domains where it relies on technical standard. This is a testable claim that should orient the policy-reform agenda toward the codification gap rather than toward generalised expansion of overlay coverage.

The Monte Carlo simulation (n=10,000) produces a probability density function for each spatial configuration. The baseline empty-land scenario, evaluated under Clause 52.32 alone, converges on a mean permittability of 27.8 per cent with a 95 per cent confidence interval of 9.3 to 46.0 per cent. The addition of a Significant Landscape Overlay (SLO) leaves the mean essentially unchanged at 27.7 per cent. The Bushfire Management Overlay reduces it to 19.8 per cent, the Environmental Significance Overlay to 18.0 per cent, and the Heritage Overlay collapses it to 2.5 per cent. A combined HO and SLO scenario converges on 2.6 per cent. The implication is that heritage friction is not additive but functionally absorptive: once a heritage trigger is in play, additional overlays do not materially change the probability surface.

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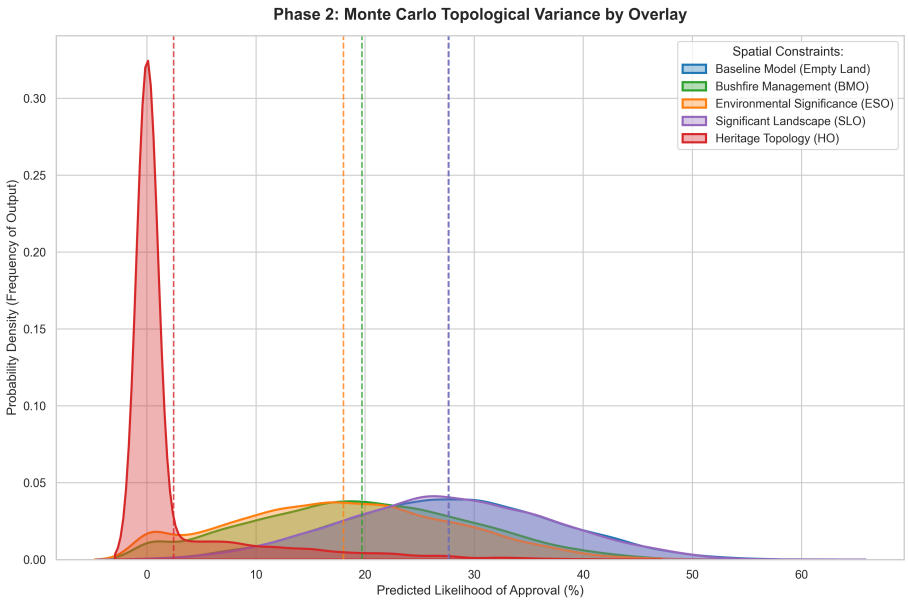


Figure 3. Phase 2 Monte Carlo probability density of permittability by overlay configuration (n=10,000 iterations per scenario). The Heritage Overlay scenario shows a sharp leftward shift and a narrow distribution, indicating high-confidence refusal. The baseline, SLO, ESO and BMO scenarios show progressively shifted but broadly overlapping distributions, indicating that the marginal effect of each overlay is partially conditional on the others.

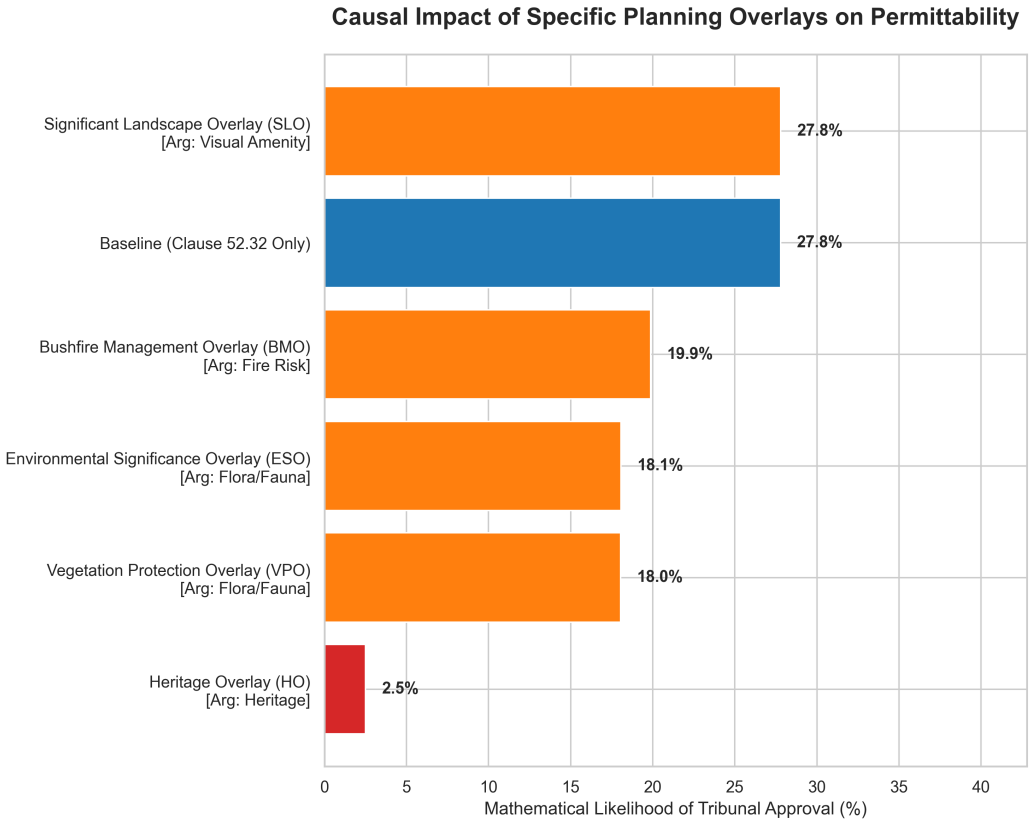


Figure 4. Causal impact of specific planning overlays on permittability. The Heritage Overlay produces a near-vetoed outcome (mean permittability 2.5 per cent), distinguishing it categorically from the other landscape and ecological controls.

The 95 per cent confidence intervals are themselves analytically important. The baseline interval (9.3 per cent to 46.0 per cent) spans nearly five-sixths of the probability range, indicating that even without overlay friction, interpretive variance in the tribunal produces enormous uncertainty for developers. This is the quantitative form of the argument Linnie (2025) makes in qualitative terms: that Victorian renewable energy planning has become an exercise in managing not the median outcome but the tail risk. The Monte Carlo output suggests that a project on unconstrained land has, depending on tribunal composition and discretionary interpretation, somewhere between a one-in-ten and a one-in-two probability of approval. That is not a sustainable basis for billion-dollar investment decisions, and it is the single most important reason that the model produces a distribution rather than a point estimate.

5.3 Phase 3: developer and regulator scenarios under OWA

The Ordered Weighted Averaging analysis transforms the Monte Carlo distribution into two stakeholder-facing scores. The developer score (75th percentile) represents the level of permittability a rational developer would assume if they believed they could mitigate average-case risks. The regulator score (25th percentile) represents the level a cautious regulator would assume if they treated every additional friction as a hard constraint.

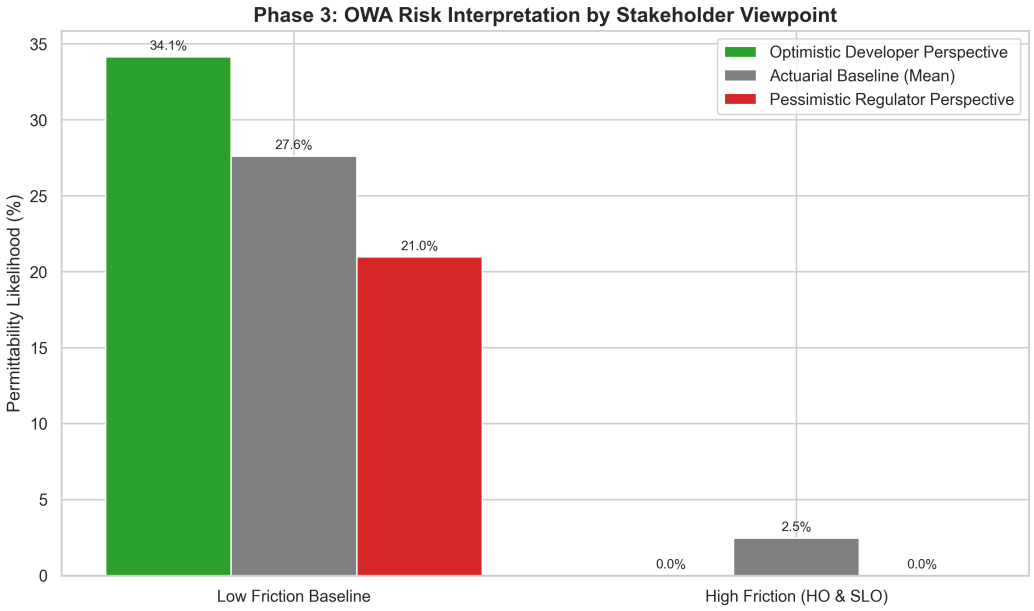


Figure 5. Phase 3 OWA risk interpretation by stakeholder viewpoint. On low-friction land the developer score (34.1 per cent) exceeds the regulator score (21.0 per cent) by 13.1 percentage points; on high-friction land with combined HO and SLO, both scores collapse to zero.

On low-friction baseline land, the developer score is 34.1 per cent and the regulator score is 21.0 per cent. The 13.1-percentage-point gap is the model's first quantification of the permittability ambiguity that has been narratively documented in the industry literature (Clean Energy Council 2025; Linnie 2025). It is the price that the planning system charges for unresolved interpretive variance, and that price is currently absorbed almost entirely by developers who must build it into discount rates and contingency budgets.

On high-friction land (combined Heritage and Significant Landscape Overlays), the developer and regulator scores both collapse to zero. The collapse is the model's most important finding: it indicates that on heritage-constrained land, the regulator-developer divergence ceases to exist because the empirical evidence forecloses optimism. Where the planning system speaks with one voice, it speaks against the project. The corollary is that the disagreement between developers and regulators is not, as commonly framed, a question of attitude or culture. It is a question of where on the spatial constraint surface the project happens to fall.

5.4 Spatial application: the Western Victorian REZ

The model was applied to a 1.5 by 1.5 degree spatial grid centred on the Western Victorian REZ, the highest-priority zone for near-term wind development. The output (Figure 1) shows a clear spatial pattern. The volcanic plains south and west of Horsham return permittability scores in the 18 to 28 per cent range, broadly consistent with the model's empty-land baseline. Linear features of low permittability follow the major drainage corridors and the heritage-listed settlements along the Western Highway. Discrete pockets of very low permittability cluster around towns and around overlay nodes that combine landscape significance with heritage protection.

Two operational observations follow. The first is that the spatial pattern is not random; it is structured by the historical geography of European settlement, which traced the same fertile volcanic soils that now host the highest wind resource. This is the empirical signature of the green-on-green paradox in Victoria. The second is that the high-permittability cells are almost always adjacent to low-permittability features. A developer who designs to the centroid of a high-permittability cell will frequently encounter overlay friction within a turbine's setback distance. The model therefore supports a design rule that prioritises cell interiors over cell edges, even where the latter offer marginally better wind resource.

Quantitative inspection of the heatmap shows clear spatial stratification across the modelled domain. Cells in the mid-permittability range (between 18 and 28 per cent) cover most of the extent and correspond to land that is investable but carries embedded interpretive risk consistent with the baseline Monte Carlo distribution. Cells above the 35 per cent threshold typically considered investable by senior debt providers cluster in two visible patches south of Donald and north-east of Stawell. Low-permittability cells (below 10 per cent) cluster along the Western Highway corridor and the Grampians margins where heritage and landscape overlays compound. The investable land in the Western Victorian REZ is therefore not absent but spatially fragmented, which has direct consequences for transmission planning: connection points sited within contiguous low-permittability clusters cannot service nearby generation projects at the rate originally anticipated in the Integrated System Plan.

5.4.1 Sensitivity to the hybrid weighting rule

The 80/20 hybrid weighting between the machine-learned coefficient and the AHP expert prior is a deliberate analytical choice rather than a derived parameter. To test whether the headline findings depend on this choice, the Monte Carlo simulation was rerun across four alternative weighting schemes. Under a 50/50 split, the HO scenario mean permittability falls from 2.5 per cent to 1.3 per cent; under a pure-ML configuration it rises to 3.6 per cent; under a pure-AHP configuration it falls to 0.5 per cent; under a 70/30 split it sits at 2.0 per cent. The structural finding — that the Heritage Overlay collapses permittability to single-digit values — is preserved across all reasonable parameter choices. The same test applied to the baseline empty-land scenario produces a wider but still informative range: 27.7 per cent under the default,

19.5 per cent under 50/50, 33.2 per cent under pure ML, 9.0 per cent under pure AHP, and 24.9 per cent under 70/30. The sensitivity test exposes a second-order finding: the AHP expert prior is doing meaningful analytical work that pure data cannot replicate, because the AHP weights for noise impact and fire risk are non-zero where the ML coefficients for those themes are zero. The hybrid rule therefore captures a layer of friction that a purely empirical model would miss. This is consistent with Kitchin's (2014) argument that computational models perform best when they explicitly preserve the expert judgement that is otherwise absorbed and obscured.

5.5 Discussion: integrating findings with the literature

Three integrating points connect the findings to the theoretical frame established in Section 3. First, the model gives quantitative form to the patchwork thesis advanced by Taylor and Hurley (2016) and McGuirk et al. (2014). The 13-percentage-point developer-regulator gap on low-friction land is the interpretive patchwork made measurable, and it indicates that regulatory drift is a hard probabilistic problem rather than a soft cultural one. Second, the empirical predominance of heritage as a refusal driver supports Porter's (2017) argument that Australian planning has failed to integrate Indigenous People and cultural heritage perspectives effectively. The dominance is structural rather than attitudinal: the Heritage Overlay framework allows for absolute objection in a way that the Environmental Significance and Significant Landscape Overlay frameworks do not, and the absence of a proportionality test means transition-relevant projects compete in a forum that was not designed to weigh them. This is the politicised commons that Brinkley (2018) describes.

Third, the model demonstrates that the procedural-justice tradition (Wolsink 2000; Haggett 2011; Hindmarsh 2014) and the computational-planning tradition (Pamučar et al. 2026; Wang et al. 2021) are complementary rather than competing. The procedural-justice literature explains why noise, fire and traffic do not predict refusal: they are resolved through community consultation and conditions of permit. The unresolved arguments — heritage and visual amenity — are precisely those for which procedural-justice processes are weakest, because they engage cultural value that cannot be mitigated by engineering. Kitchin's (2014) caution that big-data models can obscure their political content is answered empirically: when the model is built honestly, its silences become diagnostic.

6. Recommendations

Three recommendations follow from the findings. Each is scoped, measurable, and assigned to an identifiable institutional responsibility with a target timeline. All three are intended to convert permittability from a privately absorbed risk into a publicly managed decision variable.

Recommendation 1: Establish a public REZ Permittability Dataset

The Department of Transport and Planning (DTP), in collaboration with DEECA and AEMO, should publish a machine-readable spatial dataset of permittability indicators for each declared Renewable Energy Zone within twelve months of the publication of this report. The dataset should aggregate the existing VicPlan overlay layers, the EPBC Protected Matters database, the Aboriginal Cultural Heritage Register (with appropriate redactions agreed with Traditional Owners) and the longitudinal VCAT outcome corpus into a single API. Measurable success criteria are: public release of an OGC-compliant Web Feature Service by Q2 2027; integration with the VicPlan portal; uptake by at least five major industry developers within the first year of release, measured through application telemetry. Responsibility sits with the Executive Director, Planning Information and Geographic Systems, DTP. Funding should be drawn from the Transmission Investment Framework allocations established under the State Electricity Commission.

Recommendation 2: Statutory proportionality test in declared REZs

The Minister for Planning should bring forward an amendment to Clause 52.32 and to the relevant Heritage Overlay schedules of the VPPs, introducing a structured proportionality test for heritage objections to utility-scale renewable energy proposals located within declared Renewable Energy Zones, with a target gazettal date eighteen months from the publication of this report. The test should require the responsible authority to weigh the cultural heritage impact of a proposal against the quantified climate benefit of the megawatt capacity at risk, using a published methodology drawn from the DCCEEW (2023) priority list. The test should preserve absolute protection for matters of national significance and for matters identified by Registered Aboriginal Parties under the Aboriginal Heritage Act 2006, while bringing structured visibility to the marginal cases that currently absorb a disproportionate share of refusals. Responsibility sits with the Minister for Planning, supported by the Planning Policy Strategy Branch, DTP. Measurable success criteria are: gazettal of the amendment by Q4 2027; a measurable reduction in the average tribunal-stage delay for Clause 52.32 matters from the current 543.8 days to below 365 days within thirty months of gazettal; and qualitative endorsement by the First Peoples' Assembly under the Treaty engagement framework.

Recommendation 3: Pre-application permittability triage service

The Clean Energy Council, in partnership with the Department of Transport and Planning, should fund and operationalise a pre-application permittability triage service for utility-scale renewable energy proposals within nine months of the publication of this report. The service should provide a confidential, non-binding probabilistic assessment of any proposed site, drawing on the public dataset proposed in Recommendation 1 and the RE-PRI engine documented in this report. The service should be free at the point of use for proposals exceeding 50 MW. Measurable success criteria are: triage of at least fifty proposals in the first year; a documented reduction in the share of proposals withdrawn after lodgement;

and integration of the triage output into the standard VicSmart application checklist. Responsibility for service design sits with the Director, Renewable Energy Industry Development, Clean Energy Council, with regulatory oversight by the General Manager, Statutory Planning Services, DTP. Industry contributions should be structured as a levy on grid-connection applications above the 50 MW threshold.

7. Conclusion

7.1 Synthesis

Australia's electricity transition is now bottlenecked at the planning interface. This report has argued that the bottleneck is not, in the main, a function of attitude, culture or political will; it is a function of the way regulatory and spatial frictions compound and the way that compounding is currently invisible to the decision-makers who matter most. The Renewable Energy Planning Risk Index (RE-PRI) was designed to make that compounding visible. The model produces a probabilistic permittability score for any site in the Victorian planning scheme, grounded in 141 tribunal decisions, supported by a natural-language-processed corpus of 79 judgements, weighted through the Analytical Hierarchy Process and stress-tested with a 10,000-iteration Monte Carlo simulation. Its outputs are uncomfortable but legible. The Heritage Overlay collapses permittability to 2.5 per cent. The combined HO and SLO scenario is functionally vetoed. The developer-regulator gap on low-friction land is 13.1 percentage points, and is paid by developers as risk capital.

These outputs have implications beyond the immediate task of siting renewables. They suggest that the planning system itself is performing a value sort that has not been explicitly negotiated. Heritage and visual amenity have been operationally prioritised above climate benefit, not because the Victorian community has decided so, but because the architecture of the VPPs makes that the path of least resistance. The recommendations above are designed to bring that sort into the open and to give the planning system a vocabulary for the trade-offs that the transition will continue to require.

7.2 Limitations

The most consequential limitation of the RE-PRI is structural selection bias in the underlying corpus. The 141 decisions are drawn from VCAT and the Victorian Supreme Court, and by definition describe matters that escalated beyond the responsible authority. Projects approved at council without appeal, projects refused at council without merits review, and projects negotiated to permit through extended pre-lodgement engagement are all absent. The corpus therefore over-represents worst-case adversarial encounters with the planning system and likely overstates the contested-ness of Victorian renewable approvals overall. The Monte Carlo simulation partially compensates by producing a probability distribution rather than a point estimate, but no stochastic resampling of a biased sample can recover the missing uncontested cases. A subsequent iteration should triangulate the tribunal corpus against council-level permit-register data to calibrate the worst-case skew.

Three further methodological constraints shape the empirical reliability of the model. First, the overlay-tagged subset is thin (HO n=7, ESO n=2, BMO n=1), so the Phase 1 figures should be read as indicative rather than statistically robust; formal Delphi elicitation across planning lawyers, tribunal members and industry representatives would strengthen the AHP expert prior that currently compensates. Second, the Phase 2 logistic regression uses a bag-of-words frequency model that counts keyword occurrences without

discriminating between arguments the tribunal accepted and arguments it rejected. A transformer-based legal-domain language model fine-tuned on Australian tribunal corpora would capture argument valence and offer a more sensitive measure of refusal drivers. Third, interaction effects between overlays are modelled additively rather than multiplicatively, and tribunal composition is unmodelled, although decisions before judicial members behave differently from decisions before sessional senior members.

A second cluster of limitations concerns temporal and scope boundaries. The corpus spans approximately a decade, but the Victorian planning environment is reforming rapidly: the 2024 amendments to Clause 52.32, the activation of the Victorian REZ Fund and the introduction of the Transmission Investment Framework will all shift friction coefficients in ways the historic corpus cannot anticipate. A rolling 24-month retraining window will be required once enough post-reform decisions accumulate. The model also treats permittability as a property of the Victorian planning scheme alone, isolating it from three adjacent pathways that materially influence delivery: the Commonwealth EPBC Act referral process, the AEMO grid-connection queue and the ministerial call-in regime. The current RE-PRI is therefore best understood as a planning-scheme module within a larger end-to-end risk model. Strategic objection behaviour, such as the 'phantom dwellings' documented by the Clean Energy Council (2025), is also hard to observe in advance and is not represented as a spatial constraint in the current pipeline.

A final limitation is jurisdictional and cultural. The model has been built and tested on the Victorian planning scheme. Comparable friction coefficients will need to be derived empirically for New South Wales, Queensland and South Australia, which operate under different consenting frameworks. Equally important is the treatment of Aboriginal cultural heritage. The current model represents Heritage Overlay status as a binary trigger, which is reductive both quantitatively (cultural significance is tiered and contextual) and ethically (data sovereignty considerations mean substantial portions of cultural heritage information are appropriately held outside the public record). A future iteration designed in partnership with Registered Aboriginal Parties and the First Peoples' Assembly, consistent with the Victorian Treaty engagement framework, would represent cultural heritage significance as a tiered, consent-based input rather than as an external constraint (Porter 2017).

7.3 Avenues for further research

Four lines of further work are particularly tractable: expansion of the corpus to include planning panel decisions, ministerial call-ins and council-level permit registers, which would address the selection-bias problem; co-design of a Traditional Owner-led cultural heritage module that represents significance as a tiered input; integration of the RE-PRI engine with the AEMO Integrated System Plan optimiser, so transmission investment decisions can carry permittability risk as an explicit cost variable; and comparative analysis against the offshore wind regulatory framework under the Offshore Electricity Infrastructure Act 2021, which offers an instructive counterpoint to the onshore experience. Each builds on, rather than abandons, the methodological architecture documented in this report.

7.4 Closing remarks

The deeper conclusion is one of disposition. The transition is unavoidable: the capital is available, the technology is mature, the policy targets are legislated and the engineering programmes are in train. What is missing is a planning system that can process the consequences of those facts at the pace they require. Probabilistic, evidence-based feasibility assessment is one modest contribution to closing that gap. It cannot replace the harder work of political negotiation between conservation interests, Traditional

Owners, host communities and developers, but it can make that negotiation more honest by making the stakes legible. The future of Victorian planning during the transition will depend not on whether the trade-offs are comfortable, but on whether they are visible.

-END OF REPORT-

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Acknowledgement of artificial intelligence use

In accordance with the ARCH 1411 declaration requirements, the use of artificial intelligence tools in the preparation of this report is disclosed below. Generative AI assistants (Anthropic Claude and OpenAI ChatGPT) were used for limited ideation, structural review and language editing of researcher-authored prose. AI assistants were not used to generate the empirical findings, the probabilistic model, the tribunal coding decisions or the recommendations. All Python scripts used to produce the Renewable Energy Planning Risk Index (RE-PRI) engine, the Monte Carlo simulation, the Ordered Weighted Averaging analysis and the figures presented in this report were written by the author. Outputs were verified by the author against the underlying Victorian Civil and Administrative Tribunal corpus and the Victoria Planning Provisions. The completed AI cover sheet is appended to this submission.

Appendix A. Summary of RE-PRI model parameters

Table A.1 records the empirical friction coefficients derived from the 141-case longitudinal corpus. Table A.2 records the normalised natural-language-processing causal coefficients derived from the 79 fully-coded judgements. Table A.3 records the Analytical Hierarchy Process expert prior weights used as the structural anchor for the hybrid weighting rule. The Python source for the RE-PRI engine, the Monte Carlo simulator and the OWA scenario generator is published with the report submission and is available in the File Directory accompanying this thesis.

Table A.1 Empirical friction coefficients (Phase 1)

Layer	Sample n	Refusal rate	Mean delay (days)
Baseline (full corpus)	141	30.5%	580.1
Heritage Overlay (HO)	7	57.1%	988.8
Environmental Significance Overlay (ESO)	2	0.0%	81.0
Bushfire Management Overlay (BMO)	1	0.0%	n/a
Clause 52.32 (wind energy facilities)	7	28.6%	543.8
Clause 13.10 (bushfire)	3	33.3%	115.5

Table A.2 NLP causal friction coefficients (Phase 2)

Argument theme	Normalised friction multiplier
Heritage	0.9000
Visual amenity	0.3470
Flora and fauna	0.0743

Noise impact	0.0000
Traffic	0.0000
Fire risk	0.0000
Agriculture	0.0000

Table A.3 AHP expert prior weights

Argument theme	AHP prior weight
Heritage	0.80
Flora and fauna	0.70
Fire risk	0.80
Visual amenity	0.60
Traffic	0.40
Noise impact	0.30

Table A.4 Monte Carlo permittability outputs (n=10,000 per scenario)

Scenario	Mean	95% CI low	95% CI high	Developer OWA (P75)	Regulator OWA (P25)
Baseline empty land (52.32 only)	27.8%	9.3%	46.0%	34.3%	21.2%
BMO + 52.32	19.8%	0.8%	39.4%	26.7%	12.6%
ESO + 52.32	18.0%	0.0%	38.2%	24.9%	10.7%
SLO + 52.32	27.7%	8.9%	46.2%	34.4%	20.9%
HO + 52.32	2.5%	0.0%	22.7%	0.0%	0.0%
HO + SLO + 52.32	2.6%	0.0%	22.6%	0.0%	0.0%